



FITNESSWISE

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Ever since I entered the strength training and high-performance sports industry over 33 years ago, there has been a mammoth shift in many aspects of training. One major change has been the move from training hard to training smart.

While pushing one's boundaries to the limit each and every time may make sense in the mind of a hungry athlete, few realise that throttling oneself on a regular basis is neither good for performance nor for injury avoidance. Thanks to innovative coaches, significant technological progression, and brilliant researchers, we now place greater value on data and on dosing (prescribing) training prudently for each athlete. When athletes set their workload values subjectively, this becomes one aspect of the training model. The strength and conditioning (S&C) coach then takes this information and modifies the training appropriately. **Two methods of evaluation have become prominent: Rate of Perceived Effort (RPE) and Reps in Reserve (RIR).**

RPE typically uses a 1-10 scale, much like the Visual Analogue Scale (VAS) often used in clinical settings for pain. An RPE of 1 represents very light activity, while an RPE of 10 indicates maximal effort. The less prevalent method is RIR, which involves asking the athlete how many more repetitions they could have performed in a given set — essentially, how many reps were left in reserve. Both methods allow the athlete to individually evaluate the difficulty of the effort. RPE can be used to evaluate anything from a single set to an entire workout. RIR, however, cannot be used to rate the

difficulty of a whole session, but I find it to be much easier and potentially more 'accurate' than RPE. **The 1-10 RPE scale is reasonably accurate at the extremes, but becomes infuriatingly ambiguous in the middle ranges. RIR, on the other hand, appears more precise.**

The idea of dosing sets, reps, load, volume, intensity, tempo and recovery according to the athlete seems very professional and effective; however, it is far more challenging to implement than it appears, particularly when athletes are required to assess themselves accurately using RPE and RIR. This approach makes one judicious assumption: that the athlete knows their limits. However, having trained athletes for more than a year — truly training them and pushing them — it has become obvious to me that most athletes have a poor ability to accurately rate their effort, due to a variety of factors.

We can agree that using RPE or RIR is a good idea, but if we want to use it to good effect, we need to teach athletes how to accurately evaluate their efforts. If an athlete's evaluation is inaccurate, then any modification to training based on that assessment will be proportionately erroneous. Athletes cannot accurately evaluate effort without knowing their limits, so it is important that they find and understand those limits. This task, however, comes with risk. Finding the threshold requires crossing the line; one cannot simply guess where the line lies. Unfortunately, the more frequently an athlete crosses that threshold, and the more aggressively they do so, the greater the likelihood of injury.

This leads to the logical question: how does one find their limitations

safely? Unfortunately, there is no easy answer. An athlete's limits are context-dependent. Working up to a max set of 4-6 RM in the deadlift with three minutes' rest is entirely different from doing the same with only one minute's rest. Exercise pairing also influences the threshold, making it more difficult to predict accurately. Performing six repetitions of a single-leg deadlift at 75 per cent of 1 RM combined with 10 box jumps is very different from performing the same six repetitions at the same load but paired with 10 heel raises. An athlete's limitations are also movement-related. The pain and challenge of a hard set of 4-6 squats is entirely different from performing 4-6 repetitions of biceps curls. Stress levels strongly influence an athlete's limits, and these variables change daily in line with the recovery state, sometimes even within the same workout.

The best answer to this challenging question is to use the repeat-effort method and to be patient while collating data. Patience allows the athlete to be gradually brought up to, and beyond, the threshold line. Using more sets also enables a more accurate and safer identification of that threshold.

My recommendation is to test athletes regularly. By planning tests, athletes can be better prepared for maximal efforts, and coaches can more accurately estimate where the threshold lies. This, in turn, allows athletes to be guided safely across the line without injury. By testing and regularly interpreting results, RPE and RIR recommendations become more robust, minimising guesswork and improving precision. The more accurately an athlete is dosed using RPE and RIR, the lower the probability of injury and greater the potential for dynamic performance in sport.

Success mantra: Athletes cannot accurately evaluate effort without knowing their limits, so it is important that they find and understand those limits. GETTY IMAGES